# Naresh Kumar Devulapally

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## Summary


- PhD Candidate in Computer Vision, specializing in **Diffusion Models, Multimodal Generation and World Models**, seeking full-time Research Scientist roles, starting **December 2026**.
- Strong publication record at **CVPR, ICCV, AAAI, CoLM, ACM MM**, with a proven track record of developing and productionizing AI systems across Generative AI and Multimodal Learning.


## Education

**The State University of New York at Buffalo** May 2024 – May 2027 (Expected)
Ph.D., *Computer Vision and Machine Learning* | PhD Thesis Slides Buffalo, New York, USA
Advisor: Dr. Vishnu Lokhande, Committee: Dr. Siwei Lyu, Dr. Junsong Yuan

- ***Chair's Fellowship for Research Excellence**, University at Buffalo, SUNY, 2024*
- ***CSE Graduate Teaching Award for Teaching Excellence**, University at Buffalo, SUNY, 2025*

**The State University of New York at Buffalo** August 2022 – May 2024
Master of Science in Computer Science and Engineering Buffalo, New York, USA
MS Thesis: *Towards Generalizable and Robust Multimodal ML Models*

**National Institute of Technology, Tiruchirappalli** June 2016 – June 2020
Bachelor of Technology in Mechanical Engineering Tiruchirappalli, Tamil Nadu, India

## Work Experience

**Adobe** | *Research Scientist Intern* | *Manager: Dr. John Collomosse* **May 2026 – Present**
*Content Authenticity, Audio-Video Watermarking, Multimodal AI, PyTorch* San Jose, CA, USA

- **Trained a multimodal audio-video watermarking model from scratch**, joining video and audio through learned bidirectional cross-attention at the encoder and detector, recovering a **256-bit payload** with missing-modality tolerance.
- Achieved **97.2% / 96.3% bit accuracy** on AVA and VGGSound under 24 paired real codec attacks (H.264, H.265, JPEG, AAC, MP3, resampling, noise, masking), maintaining ∼48 dB video PSNR.
- Won all 24 paired attack conditions on both datasets against VideoSeal and HierMark, by **+3.2 and +5.1 points** on attacked recovery; distributed multi-node pre-training and mid-training with mixed precision.

**Computer Vision and Machine Learning Lab** | *Research Assistant, Ph.D.* **May 2024 – Present**
*Diffusion Models, Video Generation, World Foundation Models, Vision-Language Models* Buffalo, NY, USA

- Developed a prompt-inversion framework in diffusion LLMs giving interpretable, edit-friendly prompts with a **∼10× reduction in inversion time**. *(CVPR 2026, with Adobe Research)*
- Watermarking **NVIDIA Cosmos world foundation models** in collaboration with NVIDIA: LoRA on the generation stream of Cosmos3-Nano, parameter-matched against Meta's Pixel Seal, wins **all 18 attack conditions at 5 matched quality levels**, by +10.5 points clean and +19.2 under VAE regeneration.
- Built object-level, in-generation watermarking for text-to-image latent diffusion by fine-tuning only the text token embeddings: **99% bit accuracy at 48 bits with $10^5\times$ fewer trained parameters**. *(ICCV 2025, with Adobe Research)*
- Made images **unlearnable by text-to-image personalization** with a 145K-parameter UNet that perturbs the terminal latent instead of pixels: **+8 dB PSNR over the strongest prior defence**, identity match after personalization cut from 0.78 to **0.58**, protection holding through **150 DiffPure purification steps**. *(ACM MM 2025, with Adobe Research and CVG at UMBC)*

**Visual Computing Lab** | *Research Assistant, MS Thesis* **August 2022 – May 2024**
*Multimodal Learning, Transformers, Audio-Video-Text Modeling* Buffalo, NY, USA

- Developed multimodal transformer and knowledge-distillation methods for emotion recognition across audio, video and text, handling noisy, missing and imperfectly aligned modalities.

**Carbynetech, Inc.** | *Senior Data Scientist* **August 2020 – July 2022**
*Production AI, Object Detection, Forecasting Models, Optimization* Hyderabad, India

- Deployed a real-time object detection pipeline using Flask APIs on Azure Web Services for export-pallet inspection; **achieved 94% mAP** and reduced manual inspection time by over 60%.
- Led development of Trade Promotion Optimization (TPO) product, combining demand forecasting, what-if analysis, and KPI tracking; **increased ROI by over 30%**.
- Built and deployed end-to-end TPO solution (Flask & microservices) on Azure with monitoring, scalability, and performance optimizations.

## Publications

7. Yash Shah, Abhijit Chakraborty, **Naresh Kumar Devulapally**, Vishnu Lokhande, Vivek Gupta. "OSCAR: Orchestrated Self-verification and Cross-path Refinement." *Conference on Language Modeling (CoLM)*, 2026.
6. **Naresh Kumar Devulapally**, Shruti Agarwal, Vishal Asnani, Vishnu Lokhande. "Interpretable Prompts made Edit-Friendly: Token-to-Token Similarity Reduction in dLLMs for Edit-Friendly Hard Prompt Inversion." *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026.
5. Arjun Ramesh Kaushik, **Naresh Kumar Devulapally**, Vishnu Suresh Lokhande, Nalini Ratha, Venugopal Govindaraju. "Forget Less by Learning Together through Concept Consolidation." *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2026.
4. Arjun Ramesh Kaushik, **Naresh Kumar Devulapally**, Vishnu Suresh Lokhande, Nalini Ratha, Venugopal Govindaraju. "Forget Less by Learning from Parents Through Hierarchical Relationships." *AAAI Conference on Artificial Intelligence (AAAI)*, 2026.
3. **Naresh Kumar Devulapally**, Mingzhen Huang, Vishal Asnani, Shruti Agarwal, Siwei Lyu, Vishnu Lokhande. "Your Text Encoder can be an Object-Level Watermarking Controller." *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025.
2. **Naresh Kumar Devulapally**, Shruti Agarwal, Tejas Gokhale, Vishnu Lokhande. "Latent Diffusion Unlearning: Protecting Against Unauthorized Personalization Through Trajectory Shifted Perturbations." *ACM International Conference on Multimedia (ACM MM)*, 2025.
1. **Naresh Kumar Devulapally**†, Sidharth Anand†, Sreyasee Das Bhattacharjee, Junsong Yuan. "Multi-label Emotion Analysis in Conversation via Multimodal Knowledge Distillation." *ACM International Conference on Multimedia (ACM MM)*, 2023.

### Preprints

1. Mahesh Bhosale†, **Naresh Kumar Devulapally**†, Abdul Wasi, Chau Pham, Vishnu Suresh Lokhande, David Doermann. "Score-Control for Hallucination Reduction in Diffusion Models." *Preprint*, 2026.

## Technical Skills

- **Languages & Frameworks:** Python, C++, CUDA, PyTorch (DDP/FSDP), JAX/Flax, TensorFlow, Hugging Face (Transformers, Diffusers, Accelerate, PEFT), OpenCV
- **Generative & Foundation Models:** Diffusion Models, **World Foundation Models (NVIDIA Cosmos)**, Video Generative & Video-Language Models, Latent/Stable Diffusion, CLIP, Text-to-Image/Video Generation, Image Synthesis & Editing, LLMs & Diffusion LLMs, Vision-Language Models (VLM/MLLM), Prompt Inversion, Pre-training / Post-training, Fine-tuning (LoRA/PEFT), Physical AI & World Models, Large-Scale Captioning Pipelines
- **Provenance, Trustworthy AI & Evaluation:** Invisible & Object-Level Watermarking, Content Authenticity & Provenance, C2PA, Attribution, Synthetic Media Detection, Anti-Personalization, Adversarial ML, Adversarial & Codec Robustness, Benchmarking & Evaluation, AI Safety, Hallucination Mitigation
- **Systems & Scaling:** Distributed / Multi-Node Training (Slurm), Inference Optimization, Docker, Kubernetes, AWS, GCP, Azure